# PS000109 Deep Research – Data, Systems, Hazards, Competitors

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## 1. Real exposure datasets

The best find: US government agencies publish millions of real dust and chemical measurements taken on workers, free to download. None are from India, so use them to replay realistic readings through our system and to show typical levels by job, and say so clearly.

| Dataset | What's in it | Years | Use for us |
|---|---|---|---|
| MSHA Coal Dust Samples (US mine safety agency) | Breathable dust samples from coal mines: filter weights, sample time, mine area, worker's job. Since 2016 also **real-time readings from personal dust monitors worn by miners** | 2000–now | Replay real worn-badge dust readings; see what a full shift of dust looks like |
| MSHA Personal Health Samples | Personal exposure samples from metal and non-metal mines (quarries are close to our stone and cement work), plus whether protective equipment was worn | 2000–now | Exposure levels by job; mask-use comparison |

| Dataset | What's in it | Years | Use for us |
|---|---|---|---|
| OSHA Chemical Exposure Health Data (US workplace regulator) | Lab results of samples taken during inspections: substance, result, units, personal vs area sample, industry code | Many years (researchers describe ~2 million measurements) | Typical silica and dust levels by industry (cement, ceramics, construction) |
| Construction PM dataset | PM10, PM2.5 and PM1 every 2 seconds while building a concrete block wall, from an Alphasense and a **cheap Sharp sensor side by side** | 2020 | Shows how a cheap sensor tracks a better one |
| Saudi construction heat study | 19 workers, Garmin watches (heart rate, oxygen, stress), site temperature and humidity, daily symptom check-ins, June–Oct 2023 | 2023 | Closest model to our design; data only "on reasonable request" from the authors |

Two findings that change our hardware:

- **A ping-pong-ball globe is not good enough.** A lab study found 40 mm ping-pong-ball globes matched the true radiant heat poorly ($R^2$ 0.65 vs 0.92 for the standard 150 mm globe), and in wind they read almost the same as air temperature (Scientific Reports 2022). Use a 150 mm globe: a hollow copper or metal float ball painted matte black is a common low-cost option (check size and paint).
- **Cheap dust sensors need checking in humid air.** Laser dust sensors like the PMS5003 can misread when humidity is high, and published projects still argue over the right correction (aqicn PMS5003 experiment). Kandla and Mundra are coastal and humid, so compare our badge with a reference before trusting numbers.

## 2. Government systems to plug into

We should not build a new worker registry or a new health scheme. Gujarat already runs check-up schemes and mobile vans for exactly our workers; our system decides who should use them first, based on exposure, and tracks that they did.

| System | What it is | How we connect |
|---|---|---|
| e-Shram | National register of informal workers: 31.89 crore registered (Aug 2026), each with a 12-digit UAN and occupation; shares construction workers' data with state welfare boards through APIs | Use the e-Shram UAN as the worker ID on our QR card, so the record follows the worker anywhere in India |
| ABHA / ABDM (national digital health ID) | Apps register as a Health Information Provider (sends records) or User (reads records), tested in an official sandbox first | Check-up results go into the worker's own ABHA record, which only they share: our "worker-owned record" |
| Gujarat BOCW Welfare Board (construction workers) | Total Health Check-up Scheme, Dhanvantari Arogya Rath (mobile health van), Assistance for Occupational Diseases | Book due check-ups into the board's scheme and van; flag workers eligible for disease assistance |
| Salt worker schemes (Industries Commissionerate) | Mobile medical van "for health checkup of agarias", protective kits | Send the van to the salt pans with the highest heat dose first |
| ESIC (insurance for formal workers) | Kutch was recently brought under ESI coverage after a High Court case (The Wire, Oct 2026) | Route insured port and factory workers to ESIC dispensaries for their check-ups |
| National heat-illness surveillance (NPCCHH) | Health facilities report heat illness daily, March–July (Gujarat plan) | Our anonymous heat-symptom counts can feed the district's report |

Who could pay, in order of likelihood:

1. **District office (DDO Kachchh):** a pilot, since they posted the problem.

2. **BOCW Welfare Board:** it already funds check-ups and vans for construction workers from the construction cess (a levy on building costs).
3. **Employers:** the OSH Code requires annual health check-ups for prescribed employees, so the system helps them comply.
4. **CSR funds:** large companies must spend part of their profit on social causes; Kutch has big port, power and cement operators.
5. **Industries Commissionerate:** its salt-worker welfare budget already includes health check-ups.

I could not open the scheme pages on myScheme (the site errored), so check each scheme's current rules before naming it in the pitch.

## 3. Other hazards for Kachchh

After heat and dust, glare on the salt pans and noise at ports are the next best-evidenced hazards, and both are cheap to add to the same board. The rest are better handled with a short questionnaire than a sensor.

| Hazard | Who | Evidence | How to track it | Cost |
|---|---|---|---|---|
| Sun glare and UV on white salt | Salt-pan workers | Mumbai salt workers (385 studied, 2011–14): 57.1% had eye symptoms, and pterygium rose with years of salt work (PIMR 2019); Kachchh reports of eye problems from salt glare (IDR 2025) | UV sensor (GUVA-S12SD) on the site node; log goggle use; monthly eye photo | ~₹150 |
| Noise | Port, cement, crushers | Noise-induced hearing loss is notifiable (#22). NIOSH found some iPhone sound apps within ±2 dBA, Android apps not good enough (Safety+Health) | Microphone (INMP441) on the badge; check against a calibrated iPhone app | ~₹200 |
| Whole-body vibration | Truck and crane drivers | Measured under ISO 2631 (the vibration standard) in heavy-vehicle studies (PMC) | Motion sensor (MPU6050) on the seat; pain questionnaire | ~₹150 |

| Hazard | Who | Evidence | How to track it | Cost |
|---|---|---|---|---|
| Skin damage from brine | Salt-pan workers | Skin lesions reported in Kachchh (IDR); foot numbness and spine problems in an older Gujarat study (DTE 1995) | Monthly skin/foot photo + questionnaire; boots given (yes/no) | Free |
| Fatigue and long shifts | Drivers, port workers | Not checked in this round | Shift start/end from the QR card; hours over 10 a day flagged | Free |
| Posture and back pain | Loaders, salt rakers | Not checked in this round | Standard pain questionnaire every 6 months | Free |
| No water or toilets on site | Women in salt pans | Women avoid drinking because there are no toilets (Yale e360) | Supervisor ticks "water point" and "toilet" per site; shows on the district map | Free |

The last row matters: our dehydration alerts are useless if there is nowhere to drink or use a toilet, so the dashboard should show site facilities, not only worker readings.

## 4. Competitors and research worldwide

Heat wearables exist abroad but are priced per worker per month for big US employers; in India, the main worker heat protection today is insurance that pays out after hot days, not monitoring that prevents harm.

| Product or project | What it does | Gap we fill |
|---|---|---|
| Kenzen (US) | Arm patch: heart rate, activity, skin and air temperature → estimated core temperature; alerts worker and supervisor | Monthly subscription per worker; no dust; no link to check-ups |
| SlateSafety BAND V2 (US) | Arm band heat-exposure monitor with a web portal | Same: costly, heat only |
| OSHA-NIOSH Heat Safety Tool | Free app: heat index, hourly forecast, precautions, first aid; English and Spanish | Uses heat index, not WBGT; no work–rest plan; no Indian languages; no longer being updated |

| Product or project | What it does | Gap we fill |
|---|---|---|
| SEWA heat micro-insurance (India) | Pays women workers when heat crosses set thresholds: ~50,000 women in Gujarat, Maharashtra and Rajasthan in 2024, nearly ₹5 crore paid | Pays after the heat; our site readings could help trigger fairer, site-level payouts |
| Saudi construction study (research) | Watches + site sensors + daily symptom check-ins; model flagged heat risk with ~95% accuracy | 19 workers only; not a product |

Two research findings worth using:

- **Core temperature from heart rate works in the lab.** An Italian 2024 study estimated core temperature from heart rate with an error of about 0.3 °C in 13 workers in a hot chamber, similar to the US Army (Buller) method (Frontiers 2024). It used a chest strap, not a cheap band, and was not tested outdoors, so treat band-based estimates as experimental.
- **Indian workers lack water and toilets more than gadgets.** In a Heat Watch / TISS survey of 115 garment workers, 87% had heat symptoms, 73% lacked running water and 45% showed dark urine (Newslaundry 2026). Heat illness is also not recognised as an occupational disease under the ESI Act.

## 5. What to add to our build, in order

The research changes three things now and adds three things for later.

Change now (before the synopsis):

1. **Swap the ping-pong-ball globe for a 150 mm black globe.** Small globes are too inaccurate for WBGT.
2. **Use the e-Shram UAN as the worker ID** on the QR card, and say results go into the worker's ABHA record.
3. **Show existing government schemes as the delivery channel:** BOCW health check-ups and mobile vans, the salt-worker medical van, ESIC dispensaries. We decide who goes first; they do the check-up.

Add for the finale demo:

4. **Replay real dust data:** feed MSHA personal dust monitor readings or the construction PM dataset through the badge pipeline, clearly labelled.
5. **Site facilities on the dashboard:** water point and toilet yes/no per site.

6. **UV sensor on the site node** for salt pans (~₹150), with a goggles reminder.

Show as future modules:

7. Noise on the badge, vibration on driver seats, heart-rate-based core temperature from a band, and site-level data to support heat insurance payouts like SEWA's.

Pitch line: "Kenzen charges per worker per month for a US patch. We give a Kachchh salt pan a ₹2,000 site node, plug into the schemes Gujarat already runs, and make sure the right workers reach them first."

## 6. Sources


- MSHA Coal Dust Samples data set
- MSHA Personal Health Samples data set
- OSHA Chemical Exposure Health Data – field definitions
- Construction PM dataset (Mendeley)
- Saudi construction worker heat study (arXiv)
- Globe temperature measurement study (Scientific Reports 2022)
- PMS5003 sensor experiment (aqicn)
- e-Shram five years (Organiser, Aug 2026)
- ABDM HIP/HIU guidelines
- Gujarat BOCW Total Health Check-up Scheme
- Gujarat BOCW Dhanvantari Arogya Rath
- Gujarat BOCW Assistance for Occupational Disease
- Gujarat salt worker welfare schemes (Industries Commissionerate)
- Kutch gets ESI coverage (The Wire)
- Gujarat State Action Plan on Climate Change and Human Health (NCDC)
- Mumbai salt pan workers' eye study (PIMR 2019)
- Kachchh salt pans photo essay (IDR 2025)
- Toxic salt o' the earth (Down To Earth 1995)
- NIOSH on smartphone sound apps (Safety+Health)
- Whole-body vibration in heavy-equipment drivers (PMC)
- India kidney disease and outdoor workers (Yale e360)
- Kenzen body heat monitoring
- SlateSafety BAND V2
- OSHA-NIOSH Heat Safety Tool
- Informal workers and heat protection (Newslaundry, Apr 2026)

- Core temperature from heart rate (Frontiers 2024)